Regime-Switching, Economic Disasters and "Dark Matter"

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Motivation

- A new model of rare economic disasters
- Decades-old idea: the risk of unexpected large consumption declines matters
  - Helps understand risk premium, risk-free rate, excess volatility, high put option prices etc
- But the literature has had limited success:
  1. "Dark Matter" problem of calibration
     - Quantitative implications depend on the statistical properties of time-varying disaster probability
     - But since disasters are so rare, the properties not quantifiable
  2. Disasters modeled as unexpected "nuclear strikes"
     - Risk-free yield curve downward sloping
     - Low price of call options
     - Fast recoveries very unlikely

Key Idea of This Paper

1. Think of disasters as persistent bad regimes
   - The risk of large economic declines with the possibility of reversal and fast recovery
2. Back out subjective beliefs about regime likelihood and persistence from VIX

Roadmap and Contribution

1. Novel Implications
   - Relationship between VIX and historical volatility
   - Risk-free yield curve inversions and reversions
   - New mechanism of fast price recoveries
   - *High prices of both index puts and calls (conjectural)*

2. Revisit 4 puzzles
   - Predictability of returns by P/D
   - Excess Volatility
   - Low risk-free rate puzzle
   - Risk premium puzzle

Contribution [Continued]

- First attempt at using VIX to back out subjective regime beliefs
- Closed-form solution of a regime-switching model with arbitrary regime beliefs
  - Does not rely on Markov-chain transition probabilities and facilitates usage of VIX
- *A new application of Rational Inattention theory to SDF framework (Not part of today's talk)*

Related Literature

- Some related papers:
  - Rietz (1988), Longstaff and Piazessi (2004), Barro (2006), Nowotny (2011), Gabaix (2012), Nakamura et al (2013), Wachter (2013), Gourio (2013), Barro and Ursua (2017), Welch (2018), Seo and Wachter (2019), Wachter and Zhu (2019), Ghaderi et al (2022)

Model Primitives

- Aggregate consumption C_t, aggregate dividends D_t
- Two regimes: good regime R_1 and bad regime R_2
  - In the good regime, C_t and D_t grow with low volatility
  - In the bad regime, C_t and D_t decline with high volatility
  - Exact Mathematical Statement
- Bad regimes are recessions
- Good regimes are expansions
- Disasters are outcomes of persistent or acute bad regimes
- Switching from a bad regime to a good regime is the start point of recovery

Model Primitives (Continued)

- Regimes are observable
- True regime-switching probabilities unknown and time-varying
- Representative Investor with time-separable log utility prices all assets
  - Focus today on pricing a stock market index and risk-free government bonds
- $F_{t,s}$ - time-t subjective belief of being in good regime (R_1) in period s
  - Outcome of learning about relative regime likelihood in the future
  - Can be anything within the model and will be backed out from data
  - Assumptions on $F_{t,s}$

Model Primitives (Continued)

- At the core: Regime-switching model with subjective and time-varying beliefs
  - Abstracts away from transition probabilities
  - Facilitates usage of VIX
- Constant transition probabilities imply all instances of bad regime are ex-ante-identical and have time-invariant likelihood
- Modeling time-variation in transition probabilities quantitatively challenging
  - Most likely inconsistent with VIX-based calibration
- True transition probabilities matter only to the extent they affect subjective beliefs anyway

Optimal Consumption Allocation

The representative investor maximizes lifetime utility of consumption

$$\max_{\{C_s^I\}_{s=t}^{\infty}} E_t \left[\int_t^{\infty} e^{-\delta(s-t)} \log C_s^I ds \right]$$

subject to the budget constraint

$$\underbrace{W_t}_{Financial\ Wealth} + \underbrace{E_t \left[\int_t^{\infty} \frac{M_s}{M_t} Y_s ds \right]}_{PV\ of\ Non-Portfolio\ Income} \geq \underbrace{E_t \left[\int_t^{\infty} \frac{M_s}{M_t} C_s^I ds \right]}_{Cost\ of\ Lifetime\ Consumption} .$$

Equilibrium Conditions

Price of Default-Free Bond

Time-t price $q_{t,s}$ of a government bond maturing at time s is given by

$$q_{t,s} = e^{-(s-t)\left(\tilde{F}_{t,s} r_1 + \left(1-\tilde{F}_{t,s}\right) r_2\right)}$$

- r_1 is model-implied risk-free rate in the recession-free world
- r_2 is model-implied risk-free rate in the recession-only world
  - Exact Expression for r_i
- $\tilde{F}_{t,s}$ is subjective outlook over (t, s)
  - Weighted average of subjective good regime likelihood over disjoint segments of (t, s)
  - Depends on current time period t as well as future horizon s
  - Ranges from 0 to 1
  - $\tilde{F}_{t,s} = 1$ means perceived bad regime risk over (t, s) is zero
  - Construction of $\tilde{F}_{t,s}$
- Because $r_1 > r_2$, safe assets are more valuable when perceived bad regime risk is high

Index Price

Time-t Price P_t of a stock market index with current dividend level D_t satisfies

$$P_t = \frac{1}{\delta_t} D_t$$

- Extension of Gordon Growth formula with time-varying discount rate and time-varying expected dividend growth
- When perceived bad regime risk is high, risk free rate and expected dividend growth are low and risk premium high
  - Discount rates change modestly, but cash flow projections are much lower
  - δ_t is high and price is low

Index Price (Continued)

- In a recession-free world, $P_t = \frac{1}{\delta_1} D_t$
  - $\delta_1 = \underbrace{r_1^f}_{\text{Risk Free Rate}} + \underbrace{Cov_1(C, D)}_{\text{Covariance}} - \underbrace{\mu_{D,1}}_{\text{Expected } D \text{ Growth}}$
- In a recession-only world, $P_t = \frac{1}{\delta_2} D_t$
  - δ_2 analogous to δ_1, $\delta_2 \gg \delta_1$
- With horizon-invariant beliefs ($F_{t,s} = F_t \ \forall s \geq t$), all three terms replaced by time-varying probability-weighted averages
  - $\delta_t = r_t^f + Cov_t(C, D) - \mu_{D,t}$
    - $r_t^f = F_t r_1^f + (1 - F_t) r_2^f$
    - $Cov_t(C, D)$ and $\mu_{D,t}$ analogous
    - When perceived bad regime risk increases, $r_t^f + Cov_t(C, D)$ moves little but $\mu_{D,t}$ is much lower $\implies$ δ_t is higher

General expression for δ_t

General Idea

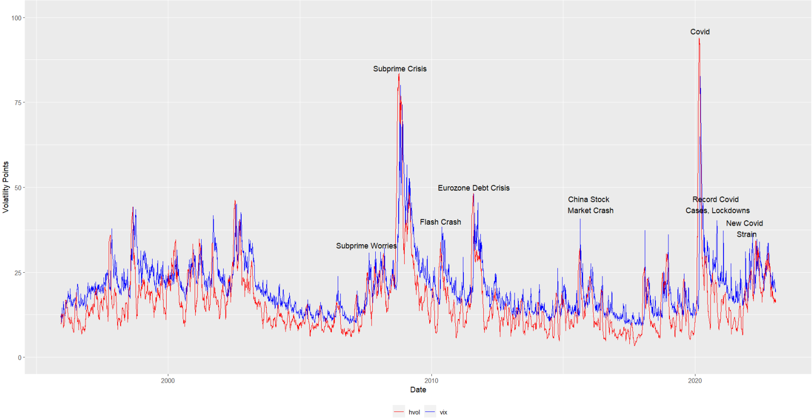
- Calibrate good regimes to non-recession data and bad regimes - to recession data
- Up to regime beliefs, all parameters depend on the moments of seasonality-adjusted real per capita consumption and dividend growth data
- Back out regime beliefs by matching the model-implied expected volatility to VIX
  - Intuition: VIX aggregates many pieces of information that is relevant to regime likelihood and is hard to represent using primitive variables alone
    - e.g. news about pandemics and political risks
  - Such information matters even if it does not always affect firm cash flows
    - Contributes to subjective understanding of macro risk

Quick Facts About VIX

- Expected volatility of S&P500 over the following 30 days
- Computed from the implied volatilities of puts and calls at various strike prices that expire in about one-month
- Measured in "volatility points": a value of 15 corresponds to annualized 15% standard deviation of S&P500 returns
- Its realized analogue: historical volatility (HVOL), computed as the annualized standard deviation of the logarithm of close-to-close daily total return
- Difference between VIX and HVOL - market "prediction error"
    - VIX forward-looking, HVOL - ex-post observed volatility

Why Use VIX?

- If high volatility is inherent to bad regimes, expectations of volatility must be informative about perceived bad regime risk
- Empirically reliable: dubbed ”fear gauge of the market”



Model-Implied Expected Volatility: Intuition

- In good regime, price volatility is low
- In bad regime, price volatility is high
- Model-implied expected volatility is the subjective-probability-weighted average of the two
- Hence, model-implied expected volatility is high when subjective bad regime likelihood is high

Backing Out Regime Beliefs: Procedure

Let $F_t(1)$ denote time-t subjective good regime likelihood at any point within the next month. Then, $F_t(1)$ is backed out from

$$F_t(1)\,\sigma_1^2 + (1 - F_t(1))\,\sigma_2^2 = \frac{(VIX/100)^2}{12}$$

- LHS: Model-implied expected variance of S&P500 returns
  - Exact Mathematical Statement
  - σ_1 volatility of S&P500 returns in good regime
  - σ_2 volatility of S&P500 returns in bad regime
- RHS: VIX converted to monthly variance measure
- Issues:
  1. Implies VIX driven exclusively by bad regime risk
  2. Implies VIX and S&P500 always move in opposite directions

Backing Out Regime Beliefs: Generalization

- Extend the model so that S&P500 has stochastic volatility (e.g. by modeling firm cash flows as having stochastic volatility)
  - Small changes in VIX can now be driven by transient shocks, but large changes must come from perceived bad regime risk
  - VIX and S&P500 can sometimes move in the same direction
- Expected Variance (EV) now weighted average of current level of variance and expected mean variance across the two regimes:

$$EV = w\sigma_t^2 + (1-w)\left(F_t(1)\,\bar{\theta}_1 + (1-F_t(1))\,\bar{\theta}_2\right)$$

Exact Mathematical Statement

Backing Out Regime Beliefs: Discussion

$F_t(1)$ backed out from

$$w\sigma_t^2 + (1-w)\left(F_t(1)\,\bar{\theta}_1 + (1-F_t(1))\,\bar{\theta}_2\right) = \frac{(VIX/100)^2}{12}$$

- σ_t^2 - Current level of variance
- $\bar{\theta}_1$ - mean variance in good regime
- $\bar{\theta}_2$ - mean variance in bad regime
- $w = \frac{1}{\bar{\kappa}}\left(1-e^{-\bar{\kappa}}\right)$, $\bar{\kappa}$ coefficient of σ_t^2 mean reversion
  - Stronger the mean-reversion, smaller the w

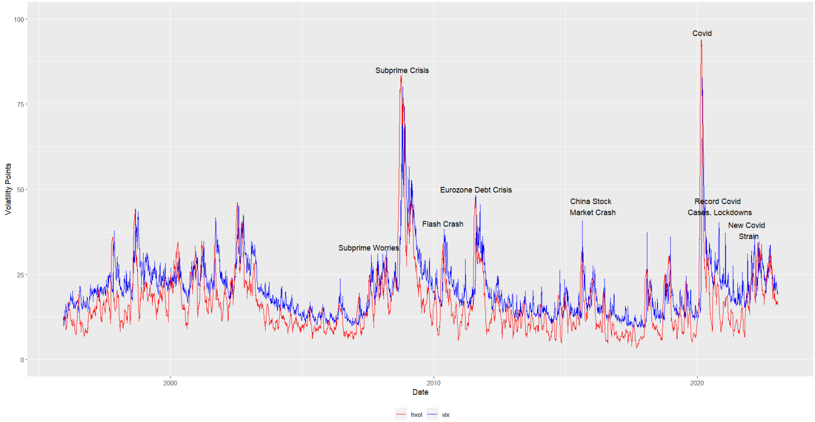
- VIX can now vary for constant $F_t(1)$
- Large changes very unlikely unless regime risk perceptions change

Going Beyond 30 Days

- VIX measures expected volatility over the following 30 days only
- Longer horizon beliefs can be calibrated using VIX futures
- Procedure is analogous: Details Here

VIX and HVOL: A Few Observations

1. During good times, VIX is higher than HVOL
2. During bad times, HVOL is higher than VIX
3. Magnitude of the discrepancy is higher during bad times



VIX and HVOL: Model Perspective

- During good times, things can only get worse
  - The possibility of regime shift is accounted for by VIX, but has no effect on HVOL unless bad news arrives Quick Proof
  - "Peso Problem" phenomenon
- During bad times, things can only get better
  - The general possibility of recovery is accounted for by VIX, but has no effect on HVOL unless recovery is in sight Quick Proof
  - Inverse "Peso Problem"
- Discrepancy is higher during bad times than during good times, if probability of recovery during bad times is higher than probability of regime shift during good times Quick Proof

Index Option Pricing: Facts and a Puzzle

1. Out-of-money observed put prices are higher than Black-Scholes prediction
   - Underprediction is stronger the further away the strike price is in the lower tail of price distribution
   - Gabaix (2012), Seo and Wachter (2019): Observed prices are higher because they account for disaster risk, which is absent in Black-Scholes world
2. Out-of-money observed call prices are higher than Black-Scholes prediction
   - Underprediction is stronger the further away the strike price is in the upper tail of price distribution
     - The pattern is more pronounced for puts than for calls
   - Puzzle: but disaster risk limits upside potential of calls and disaster risk model-implied call prices are lower than Black-Scholes prediction

Index Option Pricing: Model Perspective

- In a single-regime setup, price of a call option is given by Black-Scholes formula generalized for a dividend-paying stock
- In the recession-free world, upside potential is higher than on average $\Longrightarrow$ Call price is higher than the Black-Scholes prediction

- Conjectures:
  1. Because good regimes create excess "upside risk" relative to Black-Scholes world and call options insure against such risk, call prices are higher than Black-Scholes prediction
  2. Because bad regimes create excess downside risk relative to Black-Scholes world and put options insure against such risk, put prices are higher than Black-Scholes prediction
  3. Because bad regimes are more severe on average than good regimes, excess downside risk is bigger than excess upside risk $\Longrightarrow$ underprediction pattern stronger for put options

Fast Price Recovery

- In existing models, cash flows and future outlook are tied together
  - Future outlook is bad when cash flows are bad and recovers as cash flows recover
  - But during Covid, S&P 500 had regained about 50% of lost value by the time consumption was only midway through peak-to-trough path
- In this model, future outlook and cash flows affect prices via separate channels: $P_t = \frac{1}{\delta_t} D_t$
  - Current cash flows operate through D_t, while future outlook - through δ_t
  - P_t can start increasing despite declining D_t if improving future outlook has strong enough effect through δ_t
  - Consistent with observed VIX behavior: declined from the peak of 82.69 on March 16, 2020 to 41.67 on April 9

Risk-Free Yields

- In existing models, real yield curves downward sloping
  - The risk of a "nuclear strike" event bigger over larger horizons
  - Longer-maturity riskless assets more valuable
- In this model, real yield curve
  - Is upward-sloping if investors expect things to get better
  - Is downward-sloping if investors expect things to get worse
  - Will invert/revert anytime risk perceptions change (irrespective of cash flows)
  - Will be downward-sloping in the short-run and upward-sloping in the long-run if investors are pessimistic in the short-run, but optimistic in the long-run

Mathematical Details

Excess Volatility

- Shiller (1981): Detrended S&P500 excessively volatile relative to its detrended dividends
- Puzzle in standard Lucas-tree models: model-implied prices are as volatile as dividends
- In this model, prices vary excessively relative to dividends because they reflect time-varying bad regime risk: Conditional on no regime-shift,

$$Var\left(\log \tilde{P}_t\right) = \underbrace{\sigma_D^2 t}_{Dividend\ Variance} + \underbrace{Var\left(\log \delta_t\right)}_{Variance\ Due\ to\ Changing\ Regime\ Risk} -$$

$$-\sigma_D \underbrace{Cov\left(\log \tilde{D}_t, \log \delta_t\right)}_{\leq 0} > \sigma_D^2 t = Var\left(\log \tilde{D}_t\right),$$

$\tilde{P}_t$ and $\tilde{D}_t$ detrended price and dividends.

Predictability of Returns by P/D

- Campbell and Shiller (1988): Price-dividend ratios can predict future returns
- Puzzle in standard Lucas-tree models: model-implied price-dividend ratios are constant
- The finding can be explained by time-varying bad regime risk

Predictability of Returns by P/D (Continued)

- Investors often learn about news that have a potential to cause a regime shift
- But it is difficult to differentiate ex-ante between "false alarms" and events that actually lead to a regime shift
- Hence, increase in perceived bad regime risk often not associated with cash flow effects and reverts to average levels over time
  - VIX spikes often in response to potentially persistent problems, but only a few of those spikes are related to regime shifts
- Hence, an increase in perceived bad regime risk often decreases prices without affecting cash flows
  - Forward dividend yields increase and future capital gains are high

Predictability of Returns by P/D (Continued)

- Recall $P_t = \frac{1}{\delta_t} D_t$
- One-month return: $r_{t+1} = \frac{P_{t+1}}{P_t} + \frac{D_{t+1}}{P_t} = \frac{\delta_t}{\delta_{t+1}} \frac{D_{t+1}}{D_t} + \delta_t \frac{D_{t+1}}{D_t}$
- Assume δ_t uncorrelated with dividends conditional on no regime-shift. Then, from the Econometrician's perspective:

$$E_t\left[r_{t+1}|R_1\right] = E_t\left[\frac{\delta_t}{\delta_{t+1}}|R_1\right] E_t\left[\frac{D_{t+1}}{D_t}|R_1\right] + \delta_t E_t\left[\frac{D_{t+1}}{D_t}|R_1\right]$$

- Whenever δ_t is above average, dividend yields are expected to be above average
- Because δ_t is mean reverting, $E_t\left[\frac{\delta_t}{\delta_{t+1}}|R_1\right] > 1$ and capital gains are high

Risk Premium and Risk-Free Rate

- For reasonable levels of risk aversion in Lucas tree models, risk premium is too low (Mehra and Prescott (1985)) and risk-free rate too high (Weil 1989)
- Bad regime risk makes safe assets more desirable and risky assets less valuable relative to standard models
  - Risk free rate is lower and Risk Premium is higher

Conclusion

1. I propose a regime-switching model of disasters with subjective regime beliefs
   - To address conceptual challenges of existing models
   - To mitigate the "dark matter" problem of calibration
2. Novel implications help understand:
   - Relationship between VIX and HVOL
   - Yield curve inversions and reversions
   - New mechanism of fast recoveries
   - High put and call prices (conjectural)
3. Although quantitative work is still nascent, I develop procedure for backing out regime beliefs from VIX
4. In future work:
   - Full calibration and examination of quantitative implications
   - Generalizations, e.g. to general CRRA utility
   - Rationalizing subjective beliefs using a model of Rational Inattention learning

Conclusion

Thank You!

C_t and D_t Dynamics

In good regime R_1,

$$\frac{dC_t}{C_t} = \mu_C^{(1)} dt + \sigma_C^{(1)} dB_t,$$

$$\frac{dD_{t,j}}{D_{t,j}} = \mu_{D,j}^{(1)} dt + \sigma_{D,j}^{(1)} dB_{t,j},$$

and in bad regime R_2,

$$\frac{dC_t}{C_t} = \mu_C^{(2)} dt + \sigma_C^{(2)} dB_t,$$

$$\frac{dD_{t,j}}{D_{t,j}} = \mu_{D,j}^{(2)} dt + \sigma_{D,j}^{(2)} dB_{t,j},$$

$\mu_C^{(1)}, \mu_{D,j}^{(1)} > 0$, $\mu_C^{(2)}, \mu_{D,j}^{(2)} < 0$, $\sigma_C^{(1)} < \sigma_C^{(2)}$, $\sigma_{D,j}^{(1)} < \sigma_{D,j}^{(2)}$.

Assumptions on Subjective Beliefs

1. There exists a finite partition of (t, ∞) such that $F_{t,s} = F_t(k)$ for all s inside $k-th$ set of the partition.
   - Subjective regime likelihood constant over small future time intervals
2. For any $u \in (t, s)$, $E_t[F_{u,s}] = F_{t,s}$
   - The belief the representative investor expects to have tomorrow about a fixed future point s equals his belief about s today.
3. From the perspective of agents and conditional on time t, $F_{u,u}$ is independent from C_u and D_u, $u > t$.
   - Subjectively, future beliefs uncorrelated with future consumption and dividends

Equilibrium Derivation

First Order Conditions imply

$$\frac{M_s}{M_t} = e^{-\delta(s-t)} \left(\frac{C_s^I}{C_t^I}\right)^{-1}$$

and by market clearing,

$$C_s^I = C_s \quad \forall s \geq t.$$

The price of a claim to time-s dividend payout D_s is given by

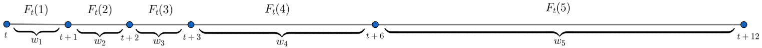
$$P_t(D_s) = E_t\left[e^{-\delta(s-t)} \left(\frac{C_s}{C_t}\right)^{-1} D_s\right]$$

Exact Expression for r_i

$$r_i = \delta + \mu_C^{(i)} - \left(\sigma_C^{(i)}\right)^2$$

- δ is a time discounting coefficient
- $\mu_C^{(i)}$ is mean consumption growth in regime R_i
- $\left(\sigma_C^{(i)}\right)^2$ is the volatility of consumption growth in regime R_i

Construction of $\tilde{F}_{t,s}$: Example



- Take 1-year horizon: $s = t + 12$
- Partition $(t, t+12)$ e.g. into 5 segments
  - Subjective good regime probability the same within segments
- $F_t(i)$ - subjective good regime probability within i-th segment
- w_i - the length of i-th segment as a fraction of $s - t$
- $$\tilde{F}_{t,s} = \sum_{i=1}^{5} w_i F_t(i)$$

General Expression for δ_t

$$\delta_t = \left(\int_t^{\infty} e^{-(s-t)\left(\tilde{F}_{t,s}\delta_1 + \left(1-\tilde{F}_{t,s}\right)\delta_2\right)} ds \right)^{-1}$$

δ_t increases whenever subjective outlook $\tilde{F}_{t,s}$ over (t, s) declines for any $s > t$.

Model-Implied Expected Variance

Suppose in regime R_i prices evolve according to

$$\frac{dP_s}{P_s} = \mu_i ds + \sigma_i dB_s, \quad i = 1, 2,$$

and assume $F_{t,s} = F_t(1)$ for all $s \in (t, t+1)$. Then,

$$E_t \left[\int_t^{t+1} \left(\frac{dP_s}{P_s} \right)^2 \right] = F_t(1)\,\sigma_1^2 + (1 - F_t(1))\,\sigma_2^2.$$

Model-Implied Generalized Expected Variance

Assume

$$\frac{dP_s}{P_s} = \mu_i ds + \sigma_s dB_s,$$

where

$$d\left(\sigma_s^2\right) = \bar{\kappa}\left(\bar{\theta}_i - \sigma_s^2\right) dt + \sqrt{\sigma_s^2}\bar{\sigma}_i dW_t,$$

$\bar{\theta}_i \equiv \frac{\bar{\sigma}_i^2}{4\bar{\kappa}}$, $i = 1, 2$, $\bar{\theta}_1 < \bar{\theta}_2$. Assume further that $F_{t,s} = F_t(1)$ for all $s \in (t, t+1)$. Then,

$$E_t\left[\int_t^{t+1}\left(\frac{dP_s}{P_s}\right)^2\right] = w\sigma_t^2 + (1-w)\left(F_t(1)\bar{\theta}_1 + (1 - F_t(1))\bar{\theta}_2\right),$$

$w = \frac{1}{\bar{\kappa}}\left(1 - e^{-\bar{\kappa}}\right).$

Going Beyond 30 Days: Details

$$wE_t\left[\sigma^2_{t+1}\right]+(1-w)\left(F_t(2)\bar{\theta}_1+(1-F_t(2))\,\bar{\theta}_2\right)=\frac{(VIX_{t+1}/100)^2}{12}$$

- $E_t\left[\sigma^2_{t+1}\right]$ - Expected level of variance after one month
- $F_t(2)$ - Bad regime probability in between months 1 and 2
- VIX_{t+1} - Expected value of VIX one month after t as implied by VIX futures

- Future periods handled similarly and recursively

VIX and HVOL: Good Times

Let $\bar{\theta}_1$ denote average variance in the good regime and let $\bar{\theta}_2$ denote average variance in the bad regime. Then, time-t subjective expected 30-day volatility (VIX_t) is

$$w\sigma_t^2 + (1-w)\left(F_t(1)\,\bar{\theta}_1 + (1-F_t(1))\,\bar{\theta}_2\right)$$

and conditional on no regime-shift, objective time-t expected 30-day volatility in good times $(E_t[HVOL\,|\,R_1])$ is

$$w\sigma_t^2 + (1-w)\,\bar{\theta}_1.$$

Hence,

$$VIX_t - E_t[HVOL\,|\,R_1] = (1-w)\times$$

$$\times \underbrace{(1-F_t(1))}_{\textit{Bad Regime Likelihood}} \times \underbrace{\left(\bar{\theta}_2 - \bar{\theta}_1\right)}_{\textit{Difference in Average Variances}} > 0.$$

VIX and HVOL: Bad Times

Let $\bar{\theta}_1$ denote average variance in the good regime and let $\bar{\theta}_2$ denote average variance in the bad regime. Then, subjective time-t expected 30-day volatility (VIX_t) is

$$w\sigma_t^2 + (1-w)\left(F_t(1)\,\bar{\theta}_1 + (1-F_t(1))\,\bar{\theta}_2\right)$$

and conditional on no regime-shift, objective time-t expected 30-day volatility in bad times $(E_t[HVOL\,|\,R_2])$ is

$$w\sigma_t^2 + (1-w)\,\bar{\theta}_2.$$

Hence,

$$VIX_t - E_t[HVOL\,|\,R_2] = (1-w)\times$$

$$\times \underbrace{F_t(1)}_{\textit{Good Regime Likelihood}} \times \underbrace{\left(\bar{\theta}_1 - \bar{\theta}_2\right)}_{\textit{Difference in Average Variances}} < 0.$$

VIX and HVOL: Discrepancy

$$\underbrace{|VIX_t - E_t\left[HVOL \,|R_1\right]|}_{\equiv M_1:\ Magnitude\ in\ Good\ Times} = (1-w)\left(1-F_t\left(1\right)\right)\left(\bar{\theta}_2 - \bar{\theta}_1\right)$$

and

$$\underbrace{|VIX_t - E_t\left[HVOL \,|R_2\right]|}_{\equiv M_2:\ Magnitude\ in\ Bad\ Times} = (1-w)\,F_t\left(1\right)\left(\bar{\theta}_2 - \bar{\theta}_1\right).$$

$E\left[M_1\right] < E\left[M_2\right]$ iff

$$\underbrace{E\left[1-F_t\left(1\right)|R_1\right]}_{Probability\ of\ Regime\ Shift\ in\ Good\ Times} < \underbrace{E\left[F_t\left(1\right)|R_2\right]}_{Probability\ of\ Recovery\ in\ Bad\ Times}.$$

Risk-Free Yields: Details

From Bond Pricing Equation, the yield $y_{t,s}$ of a risk-free bond maturing at time s is

$$y_{t,s} = \tilde{F}_{t,s} r_1 + \left(1 - \tilde{F}_{t,s}\right) r_2.$$

If s_1 and s_2 denote two future points in time, $t < s_1 < s_2$,

$$y_{t,s_2} - y_{t,s_1} = \left(\tilde{F}_{t,s_2} - \tilde{F}_{t,s_1}\right)(r_1 - r_2).$$

- Because $r_1 - r_2 > 0$, $y_{t,s_2} - y_{t,s_1} > 0$ iff $\tilde{F}_{t,s_2} - \tilde{F}_{t,s_1} > 0$
- $\tilde{F}_{t,s_2} > \tilde{F}_{t,s_1}$ means that subjective good regime likelihood is higher over time interval (s_1, s_2) relative to (t, s_1)
  - If $F_t(k)$ increasing in k, $\tilde{F}_{t,s}$ increasing in s and $y_{t,s}$ increasing in s
  - If $F_t(k)$ decreasing in k, $\tilde{F}_{t,s}$ decreasing in s and $y_{t,s}$ decreasing in s
  - If $F_t(k)$ decreasing in k for $k \leq \bar{k}$ and increasing after $\bar{k}$, $y_{t,s}$ decreasing in the short-run and increasing in the long-run